11.1 Circumference and Arc Length

Name _____ Period ____ Date _____

Geometry 11.1 · Student sheet

I can…

- find the circumference of a circle and use it to solve problems.
- find the length of an arc and use arc lengths to find measures.
- convert between degree and radian measure.

Vocabulary · fill in the missing word

Circumference — the _____ around a circle

Arc length — a portion of the _____ of a circle

Radian — the _____ of a central angle whose arc is the same length as the radius

Part 1 · Discovery: around and across

Wrap a string around each object to measure its circumference. Measure its diameter. Divide.

Object	Circumference	Diameter	Circumference ÷ diameter

Circumference ÷ diameter is always about _____.

Part 2 · Notes and examples (fill in each note, then work the example)

Circumference: $C =$ _____ $= 2\pi r$

Example 1

a. Find the circumference of a circle with a diameter of 9 centimeters.

b. A circle has a circumference of 38 feet. Find its radius.

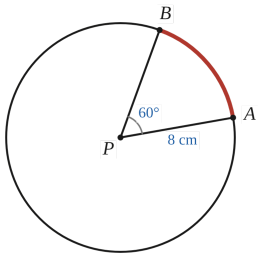
Example 2

A bike tire has a diameter of 26 inches. About how many revolutions does it make to travel 1000 feet?

Arc length: arc length of AB = (m(arc AB) ÷ 360°) · _____

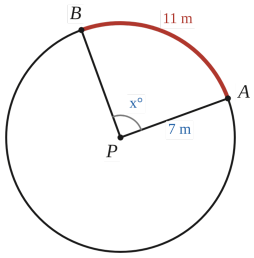
Example 3

Find the length of arc AB. Round to the nearest hundredth.



Example 4

The length of arc AB is 11 meters and the radius is 7 meters. Find m(arc AB). Round to the nearest tenth.



Radians: 180° = _____ radians. To change degrees to radians, multiply by π/180; to change radians to degrees, multiply by 180/π.

Example 5

a. Convert 45° to radians.

b. Convert $\frac{5\pi}{6}$ radians to degrees.

Part 3 · Partner practice

Partner 1

A circle has a radius of 12 inches. Find the length of a 150° arc. Round to the nearest hundredth.

Partner 2

a. Convert $\frac{3\pi}{4}$ radians to degrees.

b. Convert 120° to radians.